SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I LABHANA PRAVEEN / 192421077 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L. Praveen

Criteria	Weightage (%)	Q1 6marks	Q2 6marks	Q3 6marks	Q4 6marks	Q5 6marks
Understanding of Scenario	20%	1	0	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	2	2	1	1	2
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	0	1	0	0	0

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Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%				
Identification of Key Issues	20%				
Application of Concepts	25%				
Decision Making	20%				
Clarity of Response	15%				

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:
School of Computing → /25
School of Electronics → /26
School of Mechanical Engineering → /27
As the network engineer, design the subnet allocation by performing the following tasks:
1. Determine the network address and broadcast address for each department.

2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:
Software Development → /24
Quality Assurance → /25
Human Resources → /26
Executive Management → /27
As the network administrator, perform the following tasks:
1. Determine the network address and broadcast address for each department.
2. Identify the valid range of host IP addresses.
3. Calculate the number of usable hosts in each subnet.
4. Identify the unused IP address range remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:
2001:0db8:abcd:0000:0000:0000:0000:0000/64
As the network engineer responsible for the migration, perform the following:
1. Convert the given IPv6 address into its compressed notation.
2. Expand the compressed address back into its full hexadecimal representation.
3. Identify the network prefix using the /64 prefix length.
4. Calculate the total number of host addresses supported within the subnet.
5. Find the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
A packet arrives at the router with the destination IP address:
192.168.2.45
As the routing engineer, determine:
1. The destination subnet to which the packet belongs.
2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:
Administration LAN → 192.168.10.0/26
Finance LAN → 192.168.10.64/26

The router interfaces are configured with:
Administration Gateway – 192.168.10.1
Finance Gateway – 192.168.10.65
As the network administrator, perform the following tasks:
1. Determine the network address and broadcast address for each LAN.
2. Identify the valid host IP address range.
3. Suggest suitable IP addresses for three computers in each LAN.
4. Identify the default gateway that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Answers

1. Smart Campus Network Subnet Allocation

Given IP block: 192.168.100.0/24
- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

Step 1: Calculate subnet details

Department	CIDR	Subnet Mask	Total IPs	Usable Hosts
Computing	/25	255.255.255.128	128	126
Electronics	/26	255.255.255.192	64	62
Mechanical	/27	255.255.255.224	32	30

Step 2: Allocate subnet addresses

Allocate the subnets sequentially, starting from 192.168.100.0.

Details	Computing	Electronics	Mechanical
Network Address	192.168.100.0	192.168.100.128	192.168.100.192
First Valid Host	192.168.100.1	192.168.100.129	192.168.100.193
Last Valid Host	192.168.100.126	192.168.100.190	192.168.100.222
Broadcast Address	192.168.100.127	192.168.100.191	192.168.100.223
Usable Hosts	126	62	30

Step 3: Remaining unused IP range

The allocated subnets occupy addresses from 192.168.100.0 to 192.168.100.223.
Remaining unused range: 192.168.100.224 – 192.168.100.255
This contains 32 IP addresses, equivalent to one /27 subnet.

Conclusion: The three departments receive 126, 62, and 30 usable host addresses respectively, with 32 IP addresses remaining unused.

2. Multinational Software Company Subnet Allocation

Given IP block: 10.10.0.0/16
- Software Development → /24
- Quality Assurance → /25
- Human Resources → /26
- Executive Management → /27

Step 1: Calculate subnet details

Department	CIDR	Subnet Mask	Total IPs	Usable Hosts
Software Development	/24	255.255.255.0	256	254
Quality Assurance	/25	255.255.255.128	128	126
Human Resources	/26	255.255.255.192	64	62

Executive Management	/27	255.255.255.224	32	30

Step 2: Allocate subnet addresses

Details	Software Development	Quality Assurance	Human Resources	Executive Management
Network Address	10.10.0.0	10.10.1.0	10.10.1.128	10.10.1.192
First Valid Host	10.10.0.1	10.10.1.1	10.10.1.129	10.10.1.193
Last Valid Host	10.10.0.254	10.10.1.126	10.10.1.190	10.10.1.222
Broadcast Address	10.10.0.255	10.10.1.127	10.10.1.191	10.10.1.223
Usable Hosts	254	126	62	30

Step 3: Remaining unused IP range

The allocated subnets occupy:
- 10.10.0.0 – 10.10.0.255
- 10.10.1.0 – 10.10.1.223

Therefore, the remaining unused ranges are:
- 10.10.1.224 – 10.10.1.255
- 10.10.2.0 – 10.10.255.255

The total number of unused IP addresses is:
65,536 - (256 + 128 + 64 + 32) = 65,056

Conclusion: All four departments receive appropriately sized subnets, and 65,056 IP addresses remain available for future expansion.

3. Smart Hospital IPv6 Network

Given IPv6 network:
2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed notation

IPv6 compression rules allow leading zeros to be removed and consecutive zero blocks to be replaced with :: once.
Compressed address:
2001:db8:abcd::/64

2. Expanded full notation

The expanded address contains eight hexadecimal blocks, each with four digits.
Full notation:
2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Network prefix
The /64 prefix indicates that the first 64 bits represent the network prefix.
Network prefix:
2001:db8:abcd:0::/64

4. Host bits
IPv6 addresses contain 128 bits.
Host Bits = 128 - 64 = 64

5. Total possible host addresses
Possible Values = 2^64 = 18,446,744,073,709,551,616

Parameter	Answer
Original Address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Compressed Address	2001:db8:abcd::/64
Expanded Address	2001:0db8:abcd:0000:0000:0000:0000:0000/64
Network Prefix	2001:db8:abcd:0::/64
Host Bits	64
Total Possible Host Values	18,446,744,073,709,551,616

Conclusion: The hospital's /64 IPv6 subnet supports 2^64 possible interface identifier values, providing a very large address space for connected devices and IoT sensors.

4. ISP Routing Table and Packet Forwarding

Given routing table:
- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24
Destination IP: 192.168.2.45

1. Destination subnet
A /24 subnet uses the first three octets as the network portion.
The destination IP 192.168.2.45 belongs to:
192.168.2.0/24

2. Matching routing table entry
The matching routing table entry is:
192.168.2.0/24
This entry matches the destination address.

3. Next-hop network/interface
The router forwards the packet through the interface associated with the 192.168.2.0/24 route.
The exact interface name or next-hop IP address cannot be determined from the given information because the routing table provides only destination prefixes.

4. Default route

If no specific route matches the destination, the router can use a default route:
0.0.0.0/0
A possible configuration is:
ip route 0.0.0.0 0.0.0.0 <next-hop-IP>
The actual next-hop IP must be supplied by the network administrator.

Final answer table

Task	Answer
Destination IP	192.168.2.45
Destination Subnet	192.168.2.0/24
Matching Route	192.168.2.0/24
Forwarding Interface	Interface associated with this route; exact name unspecified
Default Route	0.0.0.0/0

Conclusion: The router selects 192.168.2.0/24 because it matches the destination IP. If no matching specific route exists, it uses the default route if one is configured.

5. Administration and Finance LAN Configuration

Given:
- Administration LAN → 192.168.10.0/26
- Finance LAN → 192.168.10.64/26
- Administration Gateway → 192.168.10.1
- Finance Gateway → 192.168.10.65

Step 1: Calculate subnet details

A /26 subnet has subnet mask 255.255.255.192.
Usable Hosts = 2^(32-26) - 2 = 2^6 - 2 = 62

Step 2: Administration LAN

- Network address: 192.168.10.0
- First valid host: 192.168.10.1
- Last valid host: 192.168.10.62
- Broadcast address: 192.168.10.63
- Default gateway: 192.168.10.1

Suggested IP addresses for three computers:
- Computer 1: 192.168.10.2
- Computer 2: 192.168.10.3
- Computer 3: 192.168.10.4

Subnet mask for all three computers: 255.255.255.192

Step 3: Finance LAN

- Network address: 192.168.10.64
- First valid host: 192.168.10.65

- Last valid host: 192.168.10.126
- Broadcast address: 192.168.10.127
- Default gateway: 192.168.10.65

Suggested IP addresses for three computers:
- Computer 1: 192.168.10.66
- Computer 2: 192.168.10.67
- Computer 3: 192.168.10.68

Subnet mask for all three computers: 255.255.255.192

Step 4: Final answer table

Details	Administration LAN	Finance LAN
Network Address	192.168.10.0	192.168.10.64
Subnet Mask	255.255.255.192	255.255.255.192
First Valid Host	192.168.10.1	192.168.10.65
Last Valid Host	192.168.10.62	192.168.10.126
Broadcast Address	192.168.10.63	192.168.10.127
Default Gateway	192.168.10.1	192.168.10.65
Computer 1	192.168.10.2	192.168.10.66
Computer 2	192.168.10.3	192.168.10.67
Computer 3	192.168.10.4	192.168.10.68
Usable Hosts	62	62